EXECUTIVE STRATEGY SERIES | ARTIFICIAL INTELLIGENCE

Organizational Transformation in the Age of AI

A research survey of how organizations transform to use AI

Prepared by **Strategy & AI Research Group** | September 1, 2026

01 | EXECUTIVE SUMMARY
The Strategic Imperative

The core philosophy that governs every argument in this document is stated upfront and without compromise:

> We do not implement AI for the organization; we solve organizational problems using AI.

Why this document exists

Artificial intelligence has moved from laboratory curiosity to strategic necessity. The macroeconomic evidence is unambiguous: the McKinsey Global Institute projects that AI could add up to **$13 trillion** to global economic output by 2030, while McKinsey's 2023 research estimates that generative AI alone could contribute **$2.6–$4.4 trillion** in annual value across 63 use cases [McKinsey Global Institute 2018; McKinsey 2023]. Yet the same body of research shows that most organizations capture almost none of this value. In the MIT Sloan–BCG survey, seven of ten companies report minimal or no impact from AI to date [MIT SMR–BCG 2019].

The gap between the promise and the outcome is not a technology gap. It is an **organizational** gap: how problems are defined, how data and talent are governed, how work is redesigned, and how the organization itself is reinvented. This document provides the framework for closing that gap.

Key findings at a glance

88%	7%	30%
Organizations use AI in at least one function [McKinsey 2025]	Report AI fully scaled across the organization [McKinsey 2025]	Gen-AI projects abandoned after proof of concept [Gartner 2023]

- **Adoption has never been higher — scaling has never been rarer.** McKinsey's 2025 survey reports that **88% of organizations** use AI in at least one business function, yet only **7%** report that AI is fully scaled across the organization [McKinsey State of AI 2025].
- **Adoption is rising faster than value.** 65% of organizations now regularly use generative AI in at least one function, up from roughly a third a year earlier [McKinsey State of AI 2024].
- **Failure is the default.** Gartner warned that through 2022, **85% of AI projects** would deliver erroneous outcomes due to bias in data, algorithms, or the teams managing them [Gartner 2018]; a further **30%** of generative-AI projects will be abandoned after proof of concept through 2025 [Gartner 2023].
- **Value follows problems, not tools.** Organizations that anchor AI to a defined business problem and a measurable KPI are the ones that scale; the rest stall at proof of concept [Gartner 2023; McKinsey 2023].
- **Measured gains exist and are concentrated.** Controlled research shows **55.8% faster** coding tasks with assistants [GitHub 2022], **+14%** customer-support productivity and **+34%** for novice agents [MIT/NBER 2023], and 25–40% faster knowledge work [BCG 2023].
- **Readiness is organizational, not technical.** The binding constraints are human capital, data capital, and organizational capital — the three "stocks" an AI-enabled organization must accumulate [MIT CISR; HBR 2018].

How to read this document

The survey is organized into five acts.

- **Act 1 · The Problem** (Parts 01–02) — the strategic imperative and the research scope.
- **Act 2 · The Landscape** (Parts 03–11) — definitions, theory, the paradigm, transformation models, the value ladder, technologies, how AI works, AI by function, and data & technology foundations.
- **Act 3 · The Organization** (Parts 12–17) — operating models, readiness, work redesign, people, capabilities and governance.
- **Act 4 · The Journey** (Parts 18–21) — why AI fails, case studies, measurement and the roadmap.
- **Act 5 · The Path Forward** (Parts 22–24) — comparative analysis, predictions and the executive conclusion.
- **Back matter** (Parts 25–28) — methodology, references, glossary and the self-assessment toolkit.

Every part ends with a **key takeaway** box; Part 28 turns the survey into a working diagnostic for your organization.

Key takeaway

Adoption is high but scaling is rare (88% use AI, 7% have scaled). The gap between the two is organizational, not technical.

02 | INTRODUCTION

From Traditional Work to AI-Based Organizations

Background of the study

The rapid maturation of generative AI, predictive systems, assistants, autonomous agents and multi-agent systems has moved artificial intelligence from a laboratory capability to an operating-model decision. Organizations everywhere are transitioning from traditional digital transformation toward AI-enabled operating models. The evidence of adoption is overwhelming:

- **88% of organizations** now use AI in at least one business function — yet only **7%** report that AI is fully scaled across the organization [McKinsey State of AI 2025].
- **78% of organizations** reported using AI in at least one function in Stanford's 2024 industry survey, while global private AI investment reached roughly **$252 billion** in 2024 [Stanford AI Index 2025].
- **64% of employees** report struggling with the time and energy to do their job — "digital debt" — and are 3.5× more likely to struggle with innovation [Microsoft Work Trend Index 2023].
- Generative AI alone could add **$2.6–$4.4 trillion** in annual value across 63 use cases [McKinsey 2023].

The research problem

There is a persistent gap between **AI experimentation** and **meaningful organizational transformation**. Organizations purchase AI tools and launch pilot projects, but they frequently fail to redesign workflows, establish governance, train employees, or measure business value. Deloitte's enterprise study — surveying **3,235 business and technology leaders across 24 countries** — documents the transition from experimentation to broader adoption as the central challenge of the current cycle [Deloitte State of AI in the Enterprise 2026]. Gartner predicts that **30% of generative-AI projects will be abandoned after proof of concept** through 2025 [Gartner 2023].

The agentic wave sharpens the problem. Many organizations introduce AI tools without answering: **which tasks should AI perform, which should it assist with, how much autonomy is acceptable, who is responsible for AI decisions, how agents connect to systems, and how unsafe or unauthorized actions are prevented** [Anthropic; NIST AI RMF]. The framework in this survey answers those questions by distinguishing assistance, partial replacement, full automation and human-only responsibility.

Research purpose

This survey develops a structured understanding of how traditional organizations can transform into **AI-assisted**, **AI-enabled**, or **AI-native** organizations. It pursues four outcomes:

1. Identify and compare the major organizational transformation models.
2. Compare real-world success and failure case studies.
3. Classify the major implementation challenges.
4. Develop a practical, phased transformation framework.

Research objectives

- Define AI-based organizational transformation and its maturity dimensions.
- Compare established transformation frameworks (McKinsey, WEF, Deloitte, Stanford, NIST, OECD).
- Examine AI use across organizational functions and measure performance change.
- Analyze successful and unsuccessful case studies with documented results.
- Identify technical, organizational, human, ethical and regulatory challenges.
- Propose a transition roadmap from traditional work to AI-assisted work.

Primary research question

How can traditional organizations transform their operating models, processes, workforce and governance to achieve sustainable value from AI?

Secondary questions include: What distinguishes AI *adoption* from AI *transformation*? Which organizational capabilities are required to scale AI? How does AI change jobs, skills and managerial responsibility? What are the most common reasons AI initiatives fail? How can organizations combine human expertise with assistants and autonomous agents? And how should AI transformation success be measured?

Key takeaway

This survey answers how traditional organizations become AI-assisted, AI-enabled or AI-native: through models, case studies and a phased roadmap.

03 | DEFINITIONS

Conceptual Foundations

Clear definitions prevent the strategy from collapsing into technology shopping. This section defines the AI systems organizations deploy and, more importantly, the four organizational states those systems produce.

Definitions of AI systems

- **Artificial intelligence** — systems performing activities associated with human intelligence: prediction, classification, language understanding, planning, generation and decision support [OECD AI Principles; EU AI Act; Stanford AI Index].
- **Generative AI** — systems that generate text, images, code, audio, video and structured data from learned patterns [Stanford AI Index; NIST Generative AI Profile].

- **AI model** — generates predictions, classifications, recommendations or content (LLMs, predictive, recommendation, classification and vision models).
- **AI assistant** — supports a person but normally does not act independently (writing, research, product-management and BI assistants).
- **AI copilot** — works inside a business process and suggests actions while the employee remains responsible (sales, coding, financial-analysis and service copilots).
- **AI agent** — receives an objective, determines the steps, uses tools and may execute actions with limited human intervention.
- **AI workflow** — a fixed, engineer-controlled sequence of AI steps; use it when the process is structured and rules are stable [Anthropic].
- **Multi-agent system** — several specialized agents cooperate (planning, retrieval, analysis, validation, reporting, governance).
- **AI-orchestrated organization** — agents coordinate information, tasks, decisions and workflows across teams and systems.

The agent vocabulary at a glance

System	Main role	Autonomy	Human role	Main risk
Assistant	Provides information	Low	User	Incorrect info
Copilot	Suggests in a process	Low–medium	Reviewer	Overreliance
Workflow	Executes fixed sequence	Fixed	Designer	Brittleness
Agent	Completes bounded tasks	Medium–high	Supervisor	Unauthorized action
Multi-agent	Coordinates specialists	Variable	Governor	Cascading failure

The central design question is not “can an agent replace a person” but **which parts of a job AI can perform safely, which parts it should assist with, and which responsibilities must remain under human judgment and accountability** [Anthropic; NIST AI RMF; Microsoft agent framework]. Anthropic’s agent research recommends **selecting the simplest architecture that reliably solves the business problem** — using predictable workflows when the process is structured, and agents only when the task requires dynamic, flexible interaction.

Adoption, enablement, transformation and AI-native

Concept	Description	Organizational signal
AI adoption	Employees or teams use individual AI tools.	Usage statistics; no process change.
AI enablement	AI is integrated into selected business processes.	Process redesign in a few functions.
AI transformation	Work, structures, capabilities and governance are redesigned around AI.	Enterprise-wide redesign, new roles, P&L impact.
AI-native	AI is the foundation of products, operations and decision-making from day one.	Business model designed around AI.

The distinction matters because the statistics are asymmetric: 88% of organizations have *adopted* AI, but only 7% have *scaled* it [McKinsey State of AI 2025]. Most organizations that call themselves “AI companies” are, by these definitions, still in adoption or early enablement. The remainder of this document is a route-map between those two states.

> **Key takeaway**
> Distinguish adoption, enablement, transformation and AI-native states, and select the simplest architecture that reliably solves the business problem.

04 | THEORY

Theoretical Foundations

Five established theories anchor the survey. Together they explain why AI transformation is an organizational problem, not a model-performance problem.

The five lenses

1 · Digital transformation theory

Organizations use digital technologies to change value creation, processes, customer relationships and operating models [MIT Sloan; HBR; Gartner; Deloitte]. AI is the newest wave of this phenomenon.

2 · Sociotechnical systems theory

AI is a joint system of technology, people, work practices, structures and rules. Performance depends on how these co-evolve — not on model quality alone [academic STS literature; HCI research].

3 · Resource-based view (RBV)

Advantage comes from valuable, rare, inimitable resources: proprietary data, AI talent, domain expertise, infrastructure, governance and trust [strategic-management literature; McKinsey; Deloitte].

4 · Dynamic capabilities

Advantage is a capacity to *sense* AI opportunities, *seize* them through investment, and *transform* structures continuously [Teece et al.; dynamic-capabilities literature].

5 · Human–AI collaboration theory

Humans and AI divide responsibility along a spectrum: AI informs, AI proposes and human approves, AI executes and human monitors, or agents execute under human governance [HBR 2018; NIST AI RMF; Microsoft; WEF].

What the theories imply

- RBV and dynamic capabilities explain **why the gap between adoption and scaling persists**: tools are commodity resources; the stocks around them are not [MIT SMR–BCG 2019].
- Sociotechnical theory explains **why AI failures are organizational**: a technically excellent model embedded in a poorly designed workflow still fails.
- Human–AI collaboration theory explains **why adoption is the bottleneck**: 30% of generative-AI projects are abandoned at the point where humans must begin working with the system [Gartner 2023].

Key takeaway

AI transformation is organizational: tools are commodity resources, but the stocks around them are not.

05 | THE AI PARADIGM

From Digitalization to AI-Enabled Reinvention

Organizations do not fail at AI because the technology is immature. They fail because they apply it with a mental model built for an earlier era. This section defines the three waves of transformation and establishes the philosophy that the remainder of this document operationalizes.

The three waves of transformation

1 · Digitalization

Converting analog records, processes and transactions into digital form. The baseline of the modern enterprise.

2 · AI Adoption

Embedding point solutions — forecasting, chatbots, vision systems — into existing workflows without redesigning them.

3 · AI-Enabled Reinvention

Redesigning business models, operating models and culture around machine intelligence. This is the frontier.

Most organizations are trapped between wave two and wave three: they have adopted AI tools, but their processes, incentives and structures were designed for a pre-AI world [McKinsey 2023]. The value does not come from the tool; it comes from the reinvention the tool enables.

The macroeconomic case

The scale of the opportunity is measurable:

- AI could add up to **$13 trillion** to global economic output by 2030 — roughly 16% of cumulative global GDP [McKinsey Global Institute 2018].
- Generative AI alone could add **$2.6–$4.4 trillion** per year across 63 use cases, and could automate 60–70% of the activities that currently consume employees' time [McKinsey 2023].
- Adoption is accelerating: 65% of organizations now regularly use generative AI in at least one function, up from roughly one-third ten months earlier, and 72% use AI in at least one business function [McKinsey State of AI 2024].

The strategic implication is blunt: the prize is real, the clock is running, and the majority of incumbents are not converting adoption into value.

Problem-first versus technology-first

The organizing philosophy of this document is:

"We do not implement AI for the organization; we solve organizational problems using AI."

The technology-first fallacy proceeds from capability to problem: "we have a large language model — where can we use it?" The problem-first discipline proceeds from pain point to capability: "which high-value organizational problem is poorly served today, and which AI capability — predictive, optimization, generative, or vision — can solve it?" The research is unequivocal about the consequences of getting this backwards: seven of ten companies report minimal or no impact from AI so far, and among organizations making significant AI investments, two in five report no bottom-line impact [MIT SMR–BCG 2019]. The survivors and winners are distinguished not by the volume of AI experiments but by the discipline with which they attach every experiment to a business problem and a measurable outcome.

Key takeaway

Solve organizational problems with AI rather than implementing AI for its own sake; value follows problem definition.

06 | TRANSFORMATION MODELS

Models for Becoming an AI-Based Organization

Six institutional models frame how organizations approach AI transformation. They differ in emphasis but converge on the same conclusion: transformation is a portfolio of strategy, work redesign, capability and governance — not a technology procurement.

Comparison of the major frameworks

Model	Core structure	Key evidence base
McKinsey	AI enablement → workflow automation → operating-model reinvention	State of AI: 88% adopt, 7% fully scaled [McKinsey 2025].
WEF / Accenture	Five transformation areas: customer experience, operations, R&D, strategy, talent	Insights from 450+ executives [WEF–Accenture 2026].
Deloitte	Maturity across strategy, leadership, technology, data, talent, governance, value, adoption	3,235 leaders across 24 countries [Deloitte 2026].
Stanford	Macro evidence: investment, performance, adoption, labor, incidents, productivity	AI Index annual reports [Stanford AI Index 2025].
NIST	Govern → Map → Measure → Manage	AI Risk Management Framework 1.0 [NIST].
OECD	Human-centered, trustworthy AI principles	OECD AI Principles [OECD].

Integrated AI transformation maturity model

L6 · AI-native operation	The organization continuously adapts around AI.
L5 · Enterprise orchestration	AI systems and agents coordinate across functions.
L4 · Workflow transformation	End-to-end processes are redesigned.
L3 · Functional integration	AI supports individual departments.
L2 · Controlled adoption	Approved tools and basic policies exist.
L1 · Unstructured experimentation	Employees use unauthorized tools.

Each level is evaluated across ten dimensions: **strategy, leadership, data, technology, processes, talent, culture, governance, measurement and customer value**. The model deliberately mirrors the ladder in Part 07 and the readiness stocks in Part 11, so the same organization can be diagnosed from three complementary angles.

Key takeaway

Use the six-level maturity model to diagnose where the organization is; the weakest stock sets the ceiling on value.

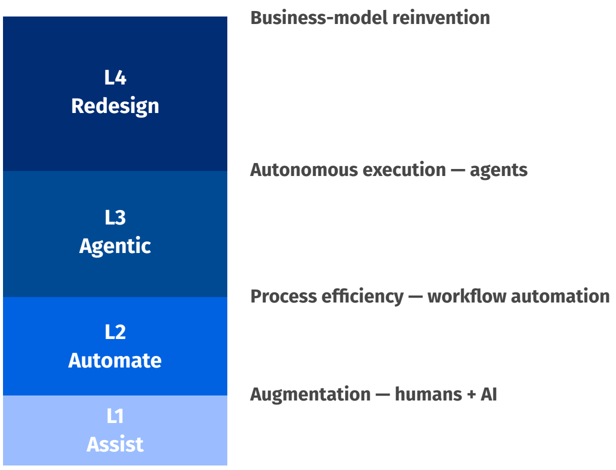
07 | THE AI VALUE CREATION LADDER

The AI Value Creation Ladder

Value from AI is not binary. It accumulates in distinct levels, and each level places different demands on the organization. Gartner's AI maturity research describes the organizational progression — from *Aware* to *Active*, *Operational*, *Systemic* and *Transformational* [Gartner AI Maturity Model] — and this section translates that progression into a concrete value ladder that CEOs can use to diagnose where their organization creates value today.

The four levels of value creation

Level	Mode	What the organization gains
1	**Assist** — augmentation	Human expertise is amplified: drafting, search, analysis, recommendations. Humans stay in control.
2	**Automate** — process efficiency	Structured workflows run without human touch: claims, onboarding, forecasting, routing.
3	**Agentic** — autonomous execution	AI agents plan and execute multi-step tasks with defined guardrails and human oversight.
4	**Redesign** — business model reinvention	Products, revenue models and value propositions are rebuilt around AI-native capabilities.



Why the levels must be sequenced

Each level assumes the one below it. Level 3 (agentic) without a Level 1 foundation of trust and a Level 2 foundation of reliable data produces autonomous failures at scale. The research quantifies how quickly this frontier is approaching: Gartner predicts that by 2028, 33% of enterprise software applications will include agentic AI — up from less than 1% in 2024 — enabling 15% of day-to-day work decisions to be made autonomously [Gartner 2024].

The diagnostic question for the executive

For every business process, ask: **where on the ladder do we operate today, and what is the next rung?** Most organizations discover that their AI portfolio is crowded at Level 1 with isolated proofs, empty at Level 2 where processes should be redesigned, and ungoverned at the frontier of Levels 3–4. The Value Creation Ladder converts that discovery into a sequenced investment agenda rather than a scramble of experiments.

Key takeaway

Sequence value creation: assist, automate, agentic, redesign — each level builds on the foundations below it.

08 | AI TECHNOLOGIES

AI Applications That Solve Organizational Problems

The correct unit of analysis is not the algorithm; it is the organizational problem. Drawing on the problem-centric categorization used in MIT Sloan research and McKinsey's use-case economics, this section groups AI applications into four categories — each defined by the class of business problem it solves [MIT SMR 2019; McKinsey 2023].

Four problem-centric categories

Predictive AI

Forecasting & risk. Demand and revenue forecasting, churn prediction, credit risk, predictive maintenance.

Optimization AI

Resource allocation. Supply-chain planning, pricing and revenue management, scheduling, routing.

Generative & Conversational AI

Knowledge & creativity. Document synthesis, semantic search, marketing content, copilots and agents.

Computer Vision

Physical-world interpretation. Visual quality inspection, safety monitoring, document capture, autonomous operations.

Where the value concentrates

McKinsey's analysis of generative AI found that roughly **75% of its potential annual value** concentrates in four functions: customer operations, marketing and sales, software engineering, and research and development [McKinsey 2023]. Predictive and optimization AI remain the most widely deployed capabilities in production, because they map directly to existing P&L line items

— revenue, cost, risk and cash. The executive implication is that a problem-first portfolio is not an even spread across all four categories; it is a concentration on the categories that map to the organization's most expensive unsolved problems.

The AI technology stack

Technology	Primary use	Autonomy	Human role	Main risk
Predictive AI	Forecasting, risk	Low	Approves	Forecast error
Classification / recommend	Segmentation, next-best-action	Low–medium	Reviews	Bias
Generative AI / LLM	Content, knowledge	Low	Verifies	Hallucination
RAG	Grounded knowledge	Medium	Validates	Stale data
Copilots / assistants	Augmented work	Low–medium	Reviewer	Overreliance
Autonomous agents	Task execution	High	Supervisor	Unauthorized action
Multi-agent BI	Coordinated analytics	High	Governor	Cascading failure

Selecting the right category

- **Uncertainty about the future → Predictive AI**.
- **Allocation of scarce resources → Optimization AI**.
- **Known answers buried in unstructured knowledge → Generative & Conversational AI**.
- **Signal trapped in images, video or physical assets → Computer Vision**.

Selection discipline is the first test of the problem-first philosophy: a technology-first organization buys the shiniest model; a problem-first organization buys the capability that retires the most expensive problem.

Seven architecture patterns

Anthropic's agent research recommends **selecting the simplest architecture that reliably solves the business problem** — fixed workflows for predictable, clearly-defined processes; agents only where dynamic interaction is genuinely required [Anthropic]. The seven patterns:

1. **Simple AI assistant** — employee → model → answer; suited to summarization, writing, translation and basic search.
2. **Retrieval-augmented assistant** — search, retrieve, generate with citations; suited to policy, documentation, support and contract analysis.
3. **Copilot workflow** — recommendation → human approval → action; suited to sales, service, product, finance and HR decisions.
4. **Fixed AI workflow** — input → steps → validation → output; suited to invoice processing, classification, standard reporting and ticket routing [Anthropic].
5. **Single AI agent** — goal → planning → tool use → result → evaluation; suited to research, IT troubleshooting and complex document analysis.
6. **Multi-agent system** — orchestrator + specialists + validation + human approval; suited to complex BI, supply-chain, R&D and cross-functional analysis.
7. **Agent-to-agent organization** — manager agent coordinates finance, sales and operations agents under human executive decision; requires strong identity, access control and validation.

The design principle: **assistants for judgment-heavy work, workflows for predictable processes, agents for bounded and measurable tasks, and multi-agent systems only when the problem genuinely requires dynamic coordination** [Anthropic; Microsoft agent framework; NIST AI RMF].

Key takeaway

Choose the AI category by the business problem (predictive, optimization, generative, vision) and use the simplest architecture pattern.

09 | HOW AI WORKS

How AI Works in Companies

AI works in companies by combining **data, models, software, employees, processes and governance**. It helps organizations understand information, predict outcomes, recommend decisions, generate content, automate repetitive work and execute controlled actions through assistants or agents. It should not be understood only as a chatbot: in an enterprise it operates at **five levels**.

Five levels of enterprise AI

- **Individual assistance** — helping one employee perform a task.
- **Functional support** — supporting sales, finance, HR or marketing.
- **Workflow automation** — coordinating several steps of an end-to-end process.
- **Organizational intelligence** — connecting information and decisions across departments.
- **Business-model transformation** — enabling new products, services and ways of working.

Deloitte's 2026 enterprise research describes the current movement from experimentation toward **scaling**, including broader workforce access and rising interest in autonomous AI agents [Deloitte State of AI in the Enterprise 2026].

AI as an employee assistant

- **Knowledge assistant** — searches documents, policies and databases; reduces search time, lowers dependency on experts and speeds onboarding.
- **Writing & communication assistant** — drafts emails, reports, meeting summaries and proposals; reduces administrative load.
- **Meeting assistant** — transcribes, extracts decisions and action items; creates searchable organizational memory.
- **Personal productivity assistant** — organizes tasks, prioritizes work and prepares follow-ups.

Key control: employees must not treat AI output as automatically correct; human review remains necessary for important decisions [HBR 2018; NIST AI RMF].

The central argument

> AI empowers companies not merely by automating tasks, but by combining assistants, predictive systems, generative models, agents, human expertise and governed data into redesigned organizational workflows.

This distinction matters because the research shows the greatest value comes when organizations **redesign work and operating models**, rather than simply making existing tasks faster [McKinsey 2025; Deloitte 2026]. McKinsey reports that nearly two-thirds of surveyed organizations had not yet begun scaling AI across the enterprise — the redesign is the missing step [McKinsey State of AI 2025].

Key takeaway

AI works at five levels, from individual assistance to business-model transformation; redesign work, not just tools.

10 | AI BY FUNCTION

AI Usage and Performance Change by Function

AI's measured impact varies sharply by the type of work. The research allows a function-by-function view of both adoption and performance change.

Measured performance change by work type

Function	AI application	Measured performance change
Software engineering	Coding assistants	**55.8% faster** task completion in controlled trials [GitHub Copilot study, 2022].
Customer support	Generative-AI copilots	**+14%** issues resolved per hour; **+34%** for novice agents [MIT/NBER call-center field experiment, 2023].
Knowledge work	Copilots and assistants	**64%** of workers report digital debt; AI frees time for higher-value work [Microsoft Work Trend Index 2023].
Consulting / analysis	LLM assistance	**25–40%** faster on idea-generation tasks; **+40%** quality on creative tasks in controlled experiments [BCG 2023].
Marketing / sales	Content generation, lead scoring	**75% of gen-AI value** across customer operations, marketing/sales, engineering and R&D [McKinsey 2023].

AI use by organizational function

Customer experience

Chatbots, personalization, intent analysis, contact-center copilots, complaint classification, journey orchestration. KPI: first-contact resolution, average handling time, CSAT, churn [WEF; CRM data].

Operations & supply chain

Demand forecasting, inventory optimization, predictive maintenance, route optimization, quality control, digital twins. KPI: forecast accuracy, inventory cost, downtime [ERP/IoT data; WEF].

Finance

Fraud detection, cash-flow prediction, invoice processing, automated reporting, scenario planning. KPI: fraud loss, close time, error rate.

Human resources

Recruitment support, skills mapping, workforce planning, learning recommendations, attrition analysis. KPI: time-to-hire, attrition, internal mobility [WEF Future of Jobs; OECD].

Marketing & sales

Lead scoring, segmentation, campaign generation, dynamic pricing, account research. KPI: conversion rate, sales-cycle duration, CAC.

R&D / product

Generative design, simulation, experiment prioritization, requirements generation, product analytics. KPI: cycle time, experiment throughput.

Strategic planning

Scenario analysis, competitive intelligence, risk forecasting, capital-allocation support. KPI: forecast error, decision latency.

Product management

Customer research, discovery, prioritization, roadmap planning, documentation. KPI: time-to-insight, feature-adoption rate.

AI as an autonomous workflow agent

An agent interprets a goal, retrieves data, plans actions, uses approved tools, validates results, requests approval where necessary, executes and logs. Every agent needs defined **identity, role, permissions, tool scope, data scope, spending limits, escalation paths and logging** — Deloitte reports organizations increasingly customize agents, making identity, authorization, monitoring and human accountability essential [Deloitte 2026; Gartner 2024].

Three worked agent workflows

Customer-service agent: customer request → intent classification → history retrieval → policy check → response generation → risk and quality validation → human approval for sensitive cases → response → audit record.

Business-intelligence agent: manager question → intent agent → data-discovery agent finds trusted data → query agent builds and validates the query → analysis agent calculates results → visualization agent charts them → explanation agent writes the answer → human review for important decisions.

Product-management agent: opportunity submitted → customer-research agent analyzes feedback → market agent analyzes competitors → analytics agent reviews usage → feasibility agent identifies constraints → prioritization agent scores the opportunity → **the product manager makes the decision**.

These patterns support the professional without replacing strategic judgment [Anthropic; Microsoft agent framework].

Interpretation

The productivity statistics are the strongest evidence for the problem-first philosophy: the gains are concentrated where the problem was precisely defined (a coding task, a customer interaction) and the workflow was designed for human–AI collaboration. Where the problem is vague, the measured gains disappear — which is exactly the population captured by the 30% proof-of- concept abandonment prediction [Gartner 2023].

Key takeaway

Measured gains concentrate where the problem is precisely defined and the workflow is redesigned; adoption without redesign yields usage, not value.

11 | DATA & TECHNOLOGY

Data and Technology Foundations

AI value is capped by data readiness and architecture. Poor data quality costs the average organization **$12.9 million per year** [Gartner 2021] — the most direct quantification of the "data trust deficit."

Data readiness

Dimension	Low maturity	High maturity	KPI
Quality	Inconsistent, incomplete	Continuously monitored	Error rate, completeness
Access	Siloed	Governed data products	Access latency
Ownership	Unclear	Enterprise-assigned	Owner per dataset
Governance	Informal	Automated lineage & controls	Audit coverage

AI architecture layers

An enterprise agent architecture can be described as nine layers, from business goals to monitoring [Anthropic; Microsoft agent framework; MCP documentation]:

1. **User and business-goal layer** — employee and

customer requests, strategic goals, events and alerts.

2. **Interaction layer** — chat, voice, dashboards, business applications, collaboration platforms.
3. **Orchestration layer** — selects the model and agent, resolves required tools, parallelism, approvals, stop and escalation conditions.
4. **Agent layer** — specialized agents (planning, research, data, coding, compliance, service, finance, quality).
5. **Model layer** — foundation models, small and domain models, predictive, embedding, reranking, vision and speech models.
6. **Knowledge and data layer** — databases, warehouses, documents, knowledge bases, APIs, ERP/CRM/BI data, product analytics.
7. **Tool and action layer** — search, queries, report generation, CRM/ERP/ticketing systems, email, calendar, payment and product-management platforms.
8. **Identity and security layer** — every agent has an identity, role, permissions, approved tools, data boundaries, transaction limits, audit logs and an emergency stop.
9. **Monitoring and evaluation layer** — accuracy, cost, latency, tool usage, failed actions, human overrides, policy violations, security incidents, satisfaction and business outcomes.

Microsoft's agent framework emphasizes **human-in-the-loop workflows**: the agent pauses, sends an approval request and waits before continuing [Microsoft agent framework]. The OWASP Top 10 for LLM applications flags **excessive agency** — granting agents more permissions than a task requires — as a first-order security risk [OWASP].

Model strategy

Compare commercial foundation models, open-source models, small language models, domain models and classical ML across **accuracy, cost, latency, privacy, reliability, explainability and vendor dependence**. No single option wins; the portfolio matters.

Multi-agent integration

For agentic systems, evaluate: agent registry and capability discovery, task routing, message formats, shared context, **agent identity and inter-agent authentication**, permission boundaries, result validation and human escalation [MCP, LangGraph, AutoGen, CrewAI documentation].

Key takeaway

Data and architecture cap AI value; govern all nine layers and give every agent a defined identity, permissions and audit trail.

12 | OPERATING MODELS

How to Organize for AI

The operating model decides whether AI scales or stalls. Five archetypes recur in the research [Deloitte; McKinsey; enterprise-architecture practice].

The five operating models

Model	Main characteristics	Strengths	Weaknesses
Centralized CoE	One central AI team owns standards, platform, governance	Consistent standards; strong control	Slow, distant from business
Federated	Business units own AI; enterprise principles only	Fast, domain-close	Duplication; inconsistent governance
Hybrid	Central platform + embedded product teams	Balance of control and speed	Needs clear decision rights
Product-based	AI managed as products with owners and roadmaps	User- and value-focused	Requires mature product discipline
Agent-orchestrated	Agents coordinate work and tools under governance	Highest automation potential	Complex governance and security

The AI product operating model

The product-based model deserves emphasis because it corrects the most common structural failure: treating AI as a project. Each AI product should have defined roles:

- **Product owner** — value and roadmap.
- **Business sponsor** — budget and outcomes.
- **Technical owner** — engineering and operations.
- **Data owner** — quality and access.
- **Risk owner** — governance and compliance.
- **User group** — adoption and feedback.

Choosing the model

- **Early maturity** → centralized: control and standards come first.
- **Diversified enterprises** → federated or hybrid: domain ownership matters more than standardization.
- **Strategic AI products** → product-based: continuity beats project completion.
- **Advanced maturity** → agent-orchestrated: autonomy requires an orchestration layer with human escalation, audit logs and permission boundaries.

Agent operations team

An advanced organization adds an **agent operations capability** responsible for: agent inventory, tool permissions, prompt and policy management, performance and cost monitoring, agent evaluation, incident response and human escalation [Deloitte 2026; Microsoft agent framework].

Organizational structure and agent model

Structure	Suitable AI model	Main advantage	Main risk
Traditional hierarchy	Assistants, copilots	Easy adoption	Limited transformation
Centralized digital team	Shared AI services	Consistency	Distance from business
Federated units	Departmental agents	Domain ownership	Duplication
Hybrid product model	Enterprise AI products	Scalable, user-centered	Complex coordination
Networked organization	Multi-agent ecosystem	Fast execution	Governance complexity

Compatibility of agents with the organization

An agent should fit the organization on seven dimensions [NIST AI RMF; EU AI Act; ISO/IEC 42001]:

- **Strategy** — connected to cost, customer experience, revenue, quality, innovation, risk or resilience objectives.
- **Processes** — a defined trigger, input, reasoning, tool use, approval, execution, output, monitoring and escalation.
- **Roles** — a clear relationship with a user, supervisor, approver, data owner, product owner, risk owner and administrator.
- **Governance** — registration, risk classification, approvals, data-access rules, testing, monitoring, incident response and retirement.
- **Technology** — integration with identity, APIs, databases, ERP/CRM, BI, collaboration and workflow systems.
- **Culture** — experimentation, training, transparent communication, participation, human accountability and safe error reporting.
- **Regulation** — classification by data, customer, employee, financial, safety and legal impact under the EU AI Act risk tiers.

Key takeaway

Pick the operating model by maturity and context; agent-orchestrated models deliver the most automation but require the strongest governance.

13 | ORGANIZATIONAL READINESS

Maturity Levels and Organizational ``Stocks"

AI readiness is frequently mistaken for technical readiness. It is not. MIT CISR's research on AI-enabled organizations shows that the binding constraints on AI value creation are organizational: strategy, data, talent, governance and culture [MIT CISR]. These constraints behave like stocks: they accumulate slowly, compound over time, and cannot be rented at the point of need.

The three organizational stocks

Human Capital

AI-literate leadership, data and ML talent, and a workforce trained to work alongside machines.

Data Capital

Trusted, governed, accessible data. Poor data quality costs the average organization $12.9 million per year [Gartner 2021].

Organizational Capital

Redesigned workflows, aligned incentives, decision rights and cross-functional collaboration.

The MIT CISR readiness lens

MIT CISR research on the AI-enabled organization emphasizes that readiness is the alignment of these stocks with strategy: organizations that treat AI as an isolated IT program accumulate tools without accumulating capability [MIT CISR]. The empirical evidence supports this framing. In the MIT Sloan–BCG global study, 90% of companies have invested in AI, but fewer than two in five report business gains, and 40% of significant investors report no bottom-line impact [MIT SMR–BCG 2019]. The minority who succeed — the "pioneers" — combine strategy, organizational behavior and technology in equal measure.

Cultural readiness: humans and machines

HBR's landmark study of 1,500 companies concluded that the greatest performance gains occur when humans and machines work together, not when machines replace humans [HBR, Wilson & Daugherty 2018]. The "collaborative intelligence" model requires cultural readiness on three fronts:

- **Trust** — employees must trust the data and the model's judgment.
- **Augmentation, not substitution** — roles are redesigned around human–machine teams, protecting employment security to reduce resistance.
- **Literacy** — every manager, not only data scientists, must read and interrogate AI outputs.

Maturity stages for each stock

Stage	Human / Data / Org capital	Evidence of readiness
Foundations	Sparse	Isolated proofs, data silos, no AI governance.
Capable	Developing	Repeatable delivery, governed data, trained teams.
Differentiating	Strong	AI embedded in core workflows with measured P&L impact.
Reinvented	System-wide	Business model, products and culture are AI-native.

The executive question is not “how many models do we run” but “how mature is each stock” — because the stock that is weakest determines the ceiling on value.

Key takeaway

The binding constraints on AI value are human capital, data capital and organizational capital — not technology.

14 | WORK TRANSFORMATION

Work and Process Transformation

The unit of AI value is the redesigned task, not the model. This part classifies work, sequences process redesign, and defines human-in-the-loop design.

Task-level classification

- Tasks that can be **fully automated** — deterministic, rule-based, high-volume.
- Tasks that can be **AI-assisted** — drafting, search, analysis, recommendation.
- Tasks that require **human judgment** — ambiguous, high-stakes, ethical.
- Tasks that should **remain human-controlled** — regulatory, reputational, empathic.
- **New tasks created by AI** — prompt design, output verification, agent supervision.

Before-and-after workflow comparison

Process element	Traditional process	AI-enabled process
Information collection	Manual retrieval	Automated retrieval and synthesis
Analysis	Human performs analysis	AI prepares analysis; human validates
Decision	Human decides	AI recommends within limits; human approves
Execution	Human performs action	Agent executes; human monitors
Quality control	Manual review	Automated checks + human review
Exception handling	Escalation to manager	AI routes exceptions automatically

Process redesign sequence

1. **Remove** unnecessary steps before automating anything.
2. **Simplify** approval chains.
3. **Standardize** data and definitions.
4. **Identify** tasks suitable for AI.
5. **Define** human checkpoints and decision rights.
6. **Automate** execution where risk allows.
7. **Measure** outcomes continuously.

Human-in-the-loop design

Human must approve every action	Human reviews exceptions	Human monitors performance	AI acts within defined boundaries

The degree of autonomy rises with trust in the model, the reversibility of errors, and the strength of audit and escalation controls [NIST AI RMF].

Assistance versus task replacement

Organizations should not replace people based only on technical capability; they should redesign work by separating tasks into activities AI can automate, activities AI can assist with, and responsibilities that must remain human-led [Anthropic; Microsoft agent framework].

- **Assistance model** — AI helps when judgment, context or accountability matters (strategy, hiring, legal, medical, executive and negotiation decisions).
- **Partial replacement** — AI replaces parts of a job that are repetitive, rule-based, verifiable and low-risk (data entry, classification, transcription, standard reports).
- **Full task automation** — AI performs an entire task when it is narrowly defined, permissions are limited, results are validated and consequences of failure are bounded.
- **Human-only responsibility** — fundamental rights, employment decisions, legal accountability, safety-critical outcomes and high-value transactions remain human-led.

Task characteristic	AI role	Human role
Repetitive, low-risk	Replace task	Monitor exceptions
Structured, verifiable	Automate task	Review samples
Complex, data-rich	Assist decision	Own decision
Ambiguous, strategic	Support analysis	Decide
High-risk, regulated	Limited assistance	Approve; remain accountable
Relationship-sensitive	Prepare suggestions	Lead interaction
Safety-critical	Monitor or simulate	Control final action

The eight autonomy factors

Before granting autonomy, assess: **task repetition**, **task complexity**, **data structure and reliability**, **error cost and reversibility**, **required human judgment** (empathy, ethics, negotiation, accountability), **explainability and auditability**, **reversibility and rollback**, and **permission scope** (systems, data, actions, external communication, spend) [NIST AI RMF; OWASP].

AI autonomy levels

Level	AI capability	Human involvement	Example
0	No AI	Human performs all work	Manual reporting
1	Provides information	Human performs task	Search assistant
2	Recommends action	Human approves	Sales recommendation
3	Performs routine task	Human reviews exceptions	Invoice classification
4	Executes workflow	Human supervises	IT ticket resolution
5	Multi-agent coordination	Human governs system	Supply-chain planning
6	Continuous operation	Human controls boundaries	Low-risk autonomous ops

The levels are **not** a replacement ladder: some tasks remain at Level 2 permanently because risk or accountability makes higher autonomy inappropriate [NIST AI RMF; EU AI Act].

Key takeaway

Separate tasks AI can automate, tasks it should assist, and responsibilities that must remain human-led; escalate autonomy gradually.

15 | PEOPLE & CULTURE

People, Skills and Organizational Culture

The workforce is the adoption bottleneck. This part defines AI literacy, specialist skills, leadership capabilities and the cultural conditions that determine whether employees use the tools.

AI literacy for every employee

- Basic AI concepts and appropriate use.
- Data privacy and security.

- Prompting, verification and **hallucination awareness**.
- Responsible-use norms [WEF Future of Jobs; OECD; NIST AI RMF].

Specialist AI skills in demand

- Data scientists and ML engineers.
- AI architects and AI product managers.
- AI governance and security specialists.
- AI interaction designers and evaluation engineers.
- Agent-orchestration engineers [WEF Future of Jobs; Stanford AI Index].

Workforce productivity and skill-gap tracking

Skill dimension	Current state (typical)	Target state
AI literacy	Under 30% of employees trained	100% of knowledge workers
Prompt & verification	Ad-hoc, ungoverned	Certified, supervised
Data skills	Data team only	Distributed data literacy
Leadership AI fluency	Low board-level fluency	Executives able to interrogate AI
Agent supervision	None	Defined supervision roles

Employee adoption factors

Adoption follows the technology-acceptance literature: **perceived usefulness**, **ease of use**, **trust**, **training**, **managerial support**, **job-security concerns** and **quality of output**. The organizational evidence is sobering: while adoption is rising — 65% of organizations now regularly use generative AI [McKinsey 2024] — value is not following automatically, because adoption without redesigned work and incentives produces usage, not outcomes.

Cultural conditions for transformation

- **Experimentation** and learning from failure.
- **Data-driven decisions** with transparent metrics.
- **Cross-functional collaboration** over silos.
- **Augmentation, not substitution** — the message that protects adoption and reduces resistance [HBR 2018].
- **Responsible risk-taking** with governance guardrails.

Key takeaway

Adoption follows perceived usefulness, trust and training; frame AI as augmentation, not substitution.

16 | AI CAPABILITIES

AI-Enabled Organizational Capabilities

AI empowers a company by strengthening ten organizational capabilities, and only when seven supporting conditions are in place.

The ten capabilities

- **Faster decision-making** — less time to find, analyze, compare and recommend.
- **Better decision quality** — patterns humans miss across large datasets, given good data and transparency.
- **Increased productivity** — less repetitive work; more time for creativity, negotiation, problem-solving and strategy.
- **Lower operating costs** — fewer manual steps, errors, rework and downtimes.
- **Improved customer experience** — faster, per-

sonalized, proactive and 24-hour service.

- **Greater scalability** — more customers, transactions and data without proportional headcount.
- **Faster innovation** — research, experiments, design and prototyping accelerate.
- **Better organizational memory** — decisions and lessons remain searchable.
- **More resilient operations** — early warnings for supply, cyber, equipment and demand risk.
- **New revenue opportunities** — AI-powered products, services and intelligent subscriptions.

How AI creates value by organizational level

Level	AI activity	Example	Main benefit
Individual	Assists an employee	Drafting a report	Productivity
Team	Supports collaboration	Summarizing project data	Coordination
Department	Automates functional work	Sales forecasting	Functional efficiency
Process	Orchestrates steps	Claims processing	Cycle-time reduction
Enterprise	Connects decisions & data	Knowledge assistant	Intelligence
Ecosystem	Connects partners	Supply-chain agents	Resilience
Business model	New offerings	AI-native analytics	Revenue

Conditions required

Clear business purpose, good data, process redesign, employee adoption, AI governance, technology integration and continuous measurement. Usage alone does not demonstrate value — the company must measure both AI system performance and business outcomes [NIST AI RMF; McKinsey 2025].

The capability-to-value model

AI capability	Organizational use	Empowerment outcome
Natural-language interaction	Plain-language questions	Wider access to information
Prediction	Forecast demand or risk	Better preparation
Recommendation	Suggest next actions	Faster decisions
Generation	Create content & documents	Higher productivity
Retrieval	Find trusted knowledge	Better organizational memory
Classification	Sort documents or tickets	Lower manual effort
Automation	Execute repetitive tasks	Lower cost and cycle time
Orchestration	Coordinate systems	End-to-end improvement
Agents	Plan and execute actions	Greater scalability
Multi-agent	Coordinate specialized AI	Complex problem-solving

Final categorization

AI in companies serves **seven purposes**: employee productivity, customer value, operational efficiency, decision intelligence, innovation, organizational coordination and business transformation. The WEF framework locates these within five transformation areas — customer experience, operations, R&D, strategy and talent [WEF–Accenture 2026] — and its agentic research predicts end-to-end work redesign as enterprises rebuild for autonomous agents [WEF 2025].

Key takeaway

AI strengthens ten organizational capabilities only when purpose, data, redesign, adoption, governance and measurement are in place.

17 | GOVERNANCE

Governance, Ethics and Regulation

Governance is the system that lets the organization trust its own AI. The Stanford AI Index documents that governance frameworks are falling behind AI capability — the "governance gap" [Stanford AI Index 2025/2026].

Responsible AI principles

Fairness, transparency, explainability, privacy, security, accountability, robustness and human oversight [NIST AI RMF; OECD AI Principles; UNESCO; ISO/IEC 42001; EU AI Act].

AI risk taxonomy

- Hallucination, bias, data leakage, prompt injection.
- Model drift, cybersecurity, unauthorized actions.
- Poor explainability, overreliance, legal liability, vendor concentration.

Governance structure and roles

Role	Responsibility
Board of directors	AI strategy oversight and risk appetite
Chief AI Officer	Enterprise AI strategy and portfolio
AI risk committee	Risk classification, incident response
Legal & compliance	EU AI Act, GDPR, IP, employment law
Product / model owner	Value, risk and lifecycle
Data owner	Data quality and access
Human approver	Final decision authority and audit trail

EU regulatory context

For EU-based organizations: the **EU AI Act** classifies AI by risk (prohibited, high-risk, transparency-related, minimal), **GDPR** governs personal data, and sector regulation and cybersecurity rules apply. These are design inputs, not afterthoughts: retrofitting governance is the most common cause of the 30% post-proof-of-concept abandonment [Gartner 2023].

Agentic AI governance

Autonomous agents require additional controls: explicit tool permissions, maximum transaction limits, approval requirements, sandboxed execution, immutable audit logs, agent identity, emergency shutdown, continuous monitoring and separation of duties [NIST AI RMF; enterprise agent platforms].

Key takeaway

Governance is a design input, not an afterthought; agentic AI requires permissions, audit, oversight and EU-AI-Act classification.

18 | WHY AI FAILS

The Organizational Gap

The most important insight in the AI research literature is that failures are predictable, patterned, and organizational — not random technical accidents. Naming the failure modes is the first step to designing them out.

The evidence on failure

- Gartner warned that through 2022, **85% of AI projects would deliver erroneous outcomes** due to bias in data, algorithms, or the responsible AI teams managing them [Gartner 2018].
- Gartner further predicts that through 2025, **30% of generative AI projects will be abandoned after proof of concept** — the point where value must move from demo to production [Gartner 2023].
- Seven in ten companies report **minimal or no impact** from AI so far, and 40% of significant AI investors report no bottom-line impact [MIT SMR–BCG 2019].

The pattern is consistent: the bottleneck sits between proof of concept and production, and it is organizational.

The four failure modes

1 · The technology-first fallacy

Capability is acquired before a problem is defined. The result is a portfolio of solutions in search of problems — demos that never become P&L impact [MIT SMR–BCG 2019].

2 · The data trust deficit

Models are only as trustworthy as the data beneath them. Unmanaged data quality, silos and weak lineage produce models that no business owner will act on [Gartner 2021].

3 · Cultural resistance & low literacy

When employees fear substitution and cannot interrogate AI outputs, they bypass the system. Augmentation without literacy produces adoption failure [HBR 2018].

4 · Poor problem definition

Fuzzy problems produce fuzzy metrics. Without a defined problem, a KPI, and an owner, an AI project cannot fail visibly — and therefore cannot be fixed or scaled [McKinsey 2023].

Why the gap is widening

The agentic frontier is making the organizational gap more expensive, not less. As Gartner projects that one-third of enterprise software will include agentic AI by 2028 and 15% of day-to-day decisions will become autonomous [Gartner 2024], the cost of weak governance, untrusted data and unprepared workforces scales with the autonomy of the system. Organizations that close the organizational gap first will compound the advantage; those that do not will automate their errors faster.

Key takeaway

AI failures are predictable and organizational; prevent them with problem definition, data trust, literacy and governance.

19 | BENCHMARK CASE STUDIES

Success and Failure Stories with Measured Results

The strongest evidence for the problem-first philosophy comes from cases with measured outcomes — in both directions. This part compares documented successes and failures.

Success: Morgan Stanley – knowledge reinvention

Morgan Stanley identified an expensive organizational problem: ~50,000 financial advisors could not search the firm's research library fast enough to answer client questions in real time. In 2023 it launched *AI @ Morgan Stanley Assistant*, built on OpenAI's GPT-4, giving advisors a conversational interface to internal research [Morgan Stanley 2023; Business Insider 2023]. A Level-1 (Assist) deployment with a precise problem definition and a measurable outcome.

Success: Coca-Cola – creative and supply-chain reinvention

Coca-Cola's *Create Real Magic* generative-AI platform — built with OpenAI and Bain — attacks a content-production bottleneck across a global brand portfolio [Forbes 2023], while predictive and optimization AI support demand forecasting and supply-chain planning. In 2024 the company committed over $1 billion to a five-year cloud and generative-AI partnership with Microsoft [Reuters 2024].

Success: Klarna – customer-service automation

Klarna's AI assistant, launched in 2024, handled roughly two-thirds of customer-service chats in its first month — the equivalent workload of about **700 full-time agents** — while maintaining comparable customer satisfaction and resolution times [Klarna 2024; press coverage]. The case demonstrates Level-2 (Automate) value with a precisely scoped workflow.

Failure: Air Canada – an unreliable assistant

In 2024 a Canadian tribunal held Air Canada liable for incorrect information provided by its AI chatbot about bereavement fares — because the company was responsible for the chatbot's outputs [tribunal ruling, 2024]. The failure was not the model; it was governance: no policy defined what the assistant could claim and who owned its outputs.

Failure: IBM Watson Health – technology-first at scale

IBM invested heavily in Watson for healthcare, but the system struggled to move from demonstration to clinical reality; the business was eventually sold off in 2022 after years of losses [published analyses]. The case is the canonical example of a technology-first strategy — capability acquired before a defined, governable problem.

Failure: Zillow Offers – scaling an unvalidated model

Zillow's algorithmic home-buying business scaled a predictive model whose error rate exceeded the trading margin, producing an **$881 million write-down** and the unit's closure in 2021 [company filings; press]. Scaling a use case without validating risk and reliability is the eighth common failure pattern — here, with a billion-dollar price tag.

Synthesis: problems beat tools

Across the research base, organizations that frame AI around business problems and KPIs reach production and scale; organizations that frame AI around technology stall at proof of concept — the population captured by Gartner's 30% abandonment prediction [Gartner 2023]. Roughly three-quarters of generative-AI value sits in a handful of functions where the problems are biggest [McKinsey 2023]. **The organization that starts with the problem wins; the organization that starts with the tool waits — or loses.**

Key takeaway

Problem-first organizations scale; tool-first organizations stall or lose — Morgan Stanley and Klarna versus Zillow and IBM Watson.

20 | MEASUREMENT

Measurement and Value Realization

"What gets measured gets scaled." The research consistently shows that organizations measuring outcomes — not activity — are the ones that move from the 88% adoption figure toward the 7% scaled figure [McKinsey State of AI 2025].

Adoption metrics

Active users, frequency of use, percentage of eligible employees using AI, completed AI-assisted tasks, business units using AI, retention, satisfaction.

Productivity metrics

Time saved, cycle-time reduction, output per employee, first-time-right rate, automation rate, manual steps removed, capacity released.

Quality metrics

Accuracy, hallucination rate, error rate, human-override rate, customer satisfaction, rework, defect rate.

Business metrics

Revenue growth, cost reduction, gross margin, conversion rate, churn, customer lifetime value, risk losses avoided, innovation cycle time.

Financial ROI and investment allocation

Investment line	What it funds	Measurement
Model & infrastructure	Foundation models, GPU, platform	Cost per inference, latency
Data capability	Quality, lineage, access	Data-error rate, access latency
Work redesign	Process, role, incentive redesign	Cycle time, capacity released
Talent & literacy	Hiring, training, change management	Skill coverage, adoption rate
Governance	Risk, audit, compliance	Incident rate, audit coverage
Transformation	Program and product teams	% workflows redesigned, ROI

Transformation metrics

Percentage of workflows redesigned, percentage of AI initiatives with business owners, AI products in production, reusable platform components, AI-related roles created, reskilling completion, cross-functional adoption, and the percentage of AI investment linked to strategic objectives.

Agent evaluation

Evaluate agents across four dimensions [NIST AI RMF; ISO/IEC 42001]:

- **Technical** — accuracy, reliability, latency, cost, tool-call success, retrieval quality, hallucination rate, failure recovery, security robustness.
- **Human** — trust, satisfaction, cognitive load, time saved, ease of use, adoption, human-override rate, quality of collaboration.
- **Business** — revenue, cost, cycle time, quality, customer satisfaction, employee capacity, risk reduction, innovation speed.
- **Governance** — approval compliance, audit completeness, access-control violations, privacy incidents, policy violations, unapproved

tool calls, shutdown effectiveness.

Evaluation category	Example metrics
Capability	Task-completion rate, accuracy
Reliability	Failure rate, recovery rate
Efficiency	Cost per task, latency
Human experience	Satisfaction, trust, cognitive load
Business value	Revenue, time saved, quality
Safety	Policy violations, unauthorized actions
Governance	Audit completeness, approval compliance

Agent replacement decision process

Before replacing a human task: define the task precisely, separate it into activities, identify repetitive versus judgment activities, assess data quality and error cost, check reversibility, define human checkpoints, test on historical examples, run a controlled pilot, compare AI versus human performance, monitor in production and **expand autonomy gradually** [Anthropic; NIST AI RMF].

Assessment question	Evidence to collect
Is the task repetitive, rules clear?	Process documentation
Is input data reliable and accessible?	Data-quality assessment
Is output easy to verify?	Test set, evaluation
Is the action reversible, error cost low?	Risk analysis
Are legal risks limited?	Compliance review
Can a human intervene?	Escalation design
Will users adopt it?	Pilot, adoption survey
Can performance be measured?	KPI baseline
Recommended AI role	**Assist / partially replace / replace**

Baseline before you build: every AI product needs a measured baseline, a target KPI, and an accountable owner.

Key takeaway

Baseline before you build and measure outcomes, not activity: adoption is not value.

21 | TRANSFORMATION ROADMAP

The Strategic Reinvention Roadmap

Transformation is a change-management problem, not a technology problem. The most widely validated change framework — Kotter's eight-step model of leading change [Kotter Inc.] — maps directly onto AI-enabled reinvention. This section compresses the eight steps into five logical phases that a CEO can sequence without depending on calendar time.

Kotter's eight steps

- **1 · Create a sense of urgency** — a compelling, quantified case for why AI reinvention cannot wait.
- **2 · Build a guiding coalition** — a cross-functional team with the authority to act.
- **3 · Form a strategic vision** — a clear picture of the reinvented organization.
- **4 · Enlist a volunteer army** — mobilize the broader organization behind the vision.
- **5 · Enable action by removing barriers** — dismantle silos, data walls and legacy rules.
- **6 · Generate short-term wins** — visible, measurable successes early.
- **7 · Sustain acceleration** — build on credibility to scale across the enterprise.
- **8 · Institute change** — anchor new behaviors in culture, systems and incentives.

The five phases of AI reinvention

Phase	Core work	Kotter steps applied
1 · Diagnosis	Map the highest-value organizational problems and audit the three stocks.	1 – Urgency
2 · Cultural foundation	Literacy, trust and a guiding coalition across the C-suite and functions.	2 – Coalition; 3 – Vision
3 · Proof of value	Targeted Assist/Automate deployments with hard KPIs and visible wins.	4 – Volunteer army; 5 – Remove barriers; 6 – Short-term wins
4 · Scaling & redesign	Workflow redesign and Agentic/Optimization deployments at enterprise scale.	7 – Sustain acceleration
5 · Reinvention	Business-model reinvention anchored in culture and incentives.	8 – Institute change

The seven-phase roadmap from traditional to AI-based

Phase	Key activities	Main risk
1 · Diagnosis	Maturity and readiness assessment	Incomplete assessment
2 · Strategic alignment	AI vision, target operating model, priorities	Technology-led strategy
3 · Controlled experimentation	Limited, scored use cases	Pilot without scale
4 · Workflow redesign	Process and role redesign	Employee resistance
5 · Platform & capability	Data, model gateway, evaluation, monitoring	Inconsistent standards
6 · Scaling	Reusable patterns, AI product teams, funding	Governance gaps
7 · Operating-model reinvention	Structures, decision rights, business model	Governance complexity

Phase 1 Diagnosis	Phase 2 Alignment	Phase 3 Experiments	Phase 4 Redesign	Phase 5 Platform	Phase 6 Scaling	Phase 7 Reinvention

Sequencing principles

- **Problems first, proofs second.** Phase 1 is a diagnostic, not a procurement exercise; it identifies the problems before any tool is chosen.
- **Every phase ends with a measurable checkpoint.** Short-term wins are the fuel of Kotter's model and the antidote to the 30% proof-of-concept abandonment rate [Gartner 2023].
- **Culture is not a phase to "get through."** The second phase builds the literacy and trust that every later phase depends on [HBR 2018].

Key takeaway

Follow Kotter's change logic through seven phases; every phase ends with a measurable checkpoint.

22 | COMPARATIVE ANALYSIS

Comparing Traditional and AI-Based Organizations

This part consolidates the comparative lenses used throughout the survey into one reference matrix.

The five organizational states

Dimension	Traditional	AI-assisted	AI-enabled	AI-native
Work method	Manual	Human + AI support	Human–AI workflows	AI-centered workflows
Decisions	Hierarchical	AI-informed	AI-augmented	AI-orchestrated
Processes	Functional silos	AI added to tasks	Processes redesigned	Designed around AI
Employee role	Executes tasks	Supervises & verifies	Collaborates with AI	Manages AI systems
Governance	Corporate controls	AI-use policies	AI lifecycle governance	Continuous governance

Assistant versus agent versus multi-agent

Dimension	Assistant	Agent	Multi-agent system
Role	Provides information	Completes tasks	Coordinates tasks
Autonomy	Low	Medium–high	Variable
Tool use	Limited	Extensive	Distributed
Human role	User	Supervisor	System governor
Main risk	Incorrect info	Unauthorized action	Cascading failure

Common transformation failure patterns

Failure pattern	Symptom	Root cause	Prevention
Pilot purgatory	Many pilots, no production	No scaling plan	Define production criteria
Automating broken processes	Faster inefficient work	No redesign	Redesign before automation
Shadow AI	Unapproved tools	Weak policy	Approved tools + monitoring
No business ownership	Tech owns the problem	Weak product model	Assign business owner
No value measurement	Usage, not outcomes	Poor KPI design	Baseline and target metrics

Integrated findings

Across every comparison: **strategy, work design, capability and governance** — not model sophistication — separate the organizations that transform from those that experiment. Context determines the right operating model; principles determine whether value is captured at all.

> **Key takeaway**
>
> Strategy, work design, capability and governance separate the organizations that transform from those that merely experiment.

23 | PREDICTIONS

Predictions of AI Usage and Organizational Change

This part consolidates what leading institutions predict about **how organizations will use AI and how they will change** on the basis of generative AI — not market size. The pattern across the research is consistent: usage will keep rising, and the differentiator will be organizational change, not technology.

Predictions of AI usage in organizations

- **Scaling is the next wave.** 88% of organizations already use AI in at least one function, but only 7% report full scaling — the coming wave is enterprise-wide deployment and workflow redesign [McKinsey State of AI 2025].
- **Generative AI becomes mainstream.** 65% of organizations now regularly use generative AI in at least one function, up from roughly a third a year earlier [McKinsey State of AI 2024].
- **Agentic AI reaches the enterprise software stack.** Gartner predicts that by 2028, **33% of enterprise software applications** will include agentic AI (up from under 1% in 2024), enabling **15% of day-to-day work decisions** to be made autonomously [Gartner 2024].
- **Consolidation after the experiment.** Through 2025, **30% of generative-AI projects** will be abandoned after proof of concept as organizations separate what scales from what does not [Gartner 2023].
- **Workforce access broadens.** Deloitte's enterprise research documents the move from experimentation to scaling, with wider workforce access to AI and rising customization of autonomous agents [Deloitte State of AI in the Enterprise 2026].

Predictions of organizational change

The World Economic Forum's transformation research — based on insights from more than **450 executives** — predicts change across five areas: customer experience, operations and supply chains, R&D, strategic planning and talent. It emphasizes human accountability, end-to-end operating-model redesign, scalable talent systems, transparency and disciplined experimentation [WEF–Accenture 2026].

McKinsey's *State of Organizations* research predicts that the organizations which capture value will be those that redesign work end-to-end — distributing responsibilities between humans, agents and automation rather than adding AI to unchanged workflows [McKinsey 2025; McKinsey Symbiotic Enterprise].

Institution	Prediction	Horizon
Gartner	33% of enterprise software includes agentic AI; 15% of decisions autonomous	2028
Gartner	30% of gen-AI projects abandoned after proof of concept	2025
McKinsey	88% use AI; only 7% fully scaled — scaling becomes the priority	Current
Deloitte	From experimentation to scaling; broader workforce access	2026
WEF	End-to-end operating-model redesign across five areas	2026
Stanford	Continued adoption growth; 78% of organizations use AI in some function	2025
Microsoft	75% of knowledge workers use AI at work; digital debt persists	2024

The prediction that matters

Every institution converges on the same forecast: **the value gap is closing for organizations that redesign work, and widening for those that do not**. The organizations that will change fastest are those already treating generative AI as a problem-solving capability embedded in redesigned workflows — with governance, skills and measurement in place — rather than as a tool portfolio. This is the basis of the problem-first strategy in Part 22.

> **Key takeaway**
>
> The value gap closes for organizations that redesign work and widens for those that do not.

24 | EXECUTIVE CONCLUSION

Executive Conclusion and Call to Action

The evidence assembled in this document converges on a single strategic imperative: **the organization that treats AI as a problem-solving capability will compound value; the organization that treats AI as a tool portfolio will accumulate cost.**

The strategic imperative in numbers

$13T	**$4.4T**	**65%**
AI value by 2030 (McKinsey 2018)	Annual gen-AI value (McKinsey 2023)	Of orgs now use gen AI (McKinsey 2024)
85%	**30%**	**7/10**
Of AI projects errored (Gartner 2018)	Abandoned after PoC (Gartner 2023)	Report minimal impact (MIT SMR–BCG 2019)

The first row is the size of the prize; the second row is the probability of mismanaging it. Both are governed by the same variable: whether the organization leads with problems or with technology.

Three immediate executive asks

1. **Authorize the audit of the organizational “stocks.”** Direct a ninety-day assessment of human capital, data capital and organizational capital against the maturity stages in Section 4. The audit’s output is a readiness scorecard, not a technology roadmap [MIT CISR].
2. **Convene an AI Reinvention Council.** Establish a cross- functional coalition — strategy, operations, data, HR and finance — with the mandate to own the problem-first agenda end to end, mirroring the guiding-coalition step of Kotter’s model [Kotter Inc.].
3. **Commit to a problem-first strategy.** Require that every AI investment be filed against a named organizational problem, a measured baseline, a KPI and an accountable owner — and that no model be funded on the strength

of a technology alone [McKinsey 2023; Gartner 2023].

> We do not implement AI for the organization; we solve organizational problems using AI.

The window for action is open, but it is not open indefinitely. The economic prize is measured in trillions, the failure rate is measured in the majority, and the frontier — agentic AI — is arriving in years, not decades [Gartner 2024]. The board and the executive committee should treat this document not as a briefing but as a decision: to close the organizational gap now, or to compete from behind.

The design principle for AI agents

The strongest design principle that synthesizes this document:

> Use assistants for judgment-heavy work, workflows for predictable processes, agents for bounded and measurable tasks, and multi-agent systems only when the problem genuinely requires dynamic coordination among specialists.

Organizations should not replace people with agents based only on technical capability. They should redesign work by separating tasks into activities AI can automate, activities AI can assist with, and responsibilities that must remain human-led because they require judgment, accountability, empathy, ethical reasoning or legal responsibility [Anthropic; NIST AI RMF; Microsoft agent framework].

Key takeaway

Audit the three stocks, convene an AI Reinvention Council, and commit to a problem-first strategy.

25 | METHODOLOGY

Research Methodology and Expected Contributions

Methodology

- **Systematic literature review** — academic publications, consulting research, government frameworks, standards, company reports and regulatory publications.
- **Comparative framework analysis** — McKinsey, WEF, Deloitte, Stanford, NIST, OECD and academic models across strategy, technology, data, people, process, governance, measurement and structure.
- **Multiple-case study method** — organizations compared across industry, size and geography using a consistent template.
- **Expert interviews and employee surveys** — executives, AI leaders, product managers, engineers, compliance and HR.
- **Quantitative and qualitative analysis** — maturity, outcomes, cluster and factor analysis, thematic coding.

Evidence hierarchy

Audited financial/operational data and peer-reviewed studies rank above official company reports, which rank above vendor-published case studies and journalistic reports — and marketing claims are treated as the weakest evidence. This protects the survey from treating vendor claims as verified results.

Expected contributions

- **Theoretical** — an integrated framework connecting technology, organizational design, human work, governance and value.
- **Practical** — an AI maturity assessment, use-case prioritization model, operating-model options, workforce framework, governance checklist, case-study database and transformation roadmap.
- **Product management** — a defined role for AI product managers translating business problems into AI products with measurable adoption and outcomes.

Key takeaway

The evidence hierarchy protects this survey from vendor claims; prefer primary sources for every major claim.

26 | REFERENCES

References and Source Base

This survey is intended as a source of truth for downstream documents. The following references support the claims made throughout; each is cited inline in the relevant part. Sources are grouped by type and ordered by institution.

Institutional reports

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- Kotter Inc. *The 8-Step Process for Leading Change*.

Standards and regulation

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- OECD. *AI Principles* (2019).
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- European Union. *EU AI Act* (Regulation on Harmonised Rules on AI, 2024).
- European Data Protection Board. Guidance on generative AI and GDPR.
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- MITRE. *ATLAS* (adversarial ML threat framework).

Technical research and documentation

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- BCG Henderson Institute. Generative-AI consultant experiment (2023).
- LangChain / LangGraph, Microsoft AutoGen, CrewAI — agent and multi-agent system documentation.
- Model Context Protocol (MCP) — official technical documentation.

Company and industry sources

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- The Coca-Cola Company. *Create Real Magic* (2023); Microsoft partnership (Reuters, 2024).
- Klarna. AI assistant announcement and reported metrics (2024).
- Zillow Group. Annual reports and filings on Zillow Offers (2021).
- Air Canada. Tribunal ruling on AI-chatbot information (2024).
- IBM. Published analyses of the Watson Health business (2022).

Peer-reviewed and academic sources

- Teece, D. Dynamic capabilities and strategic management.
- Digital transformation and sociotechnical-systems literature (MIT Sloan, HCI research).
- Human–AI collaboration and algorithmic-management research (Oxford Internet Institute, Cambridge CFI).
- AI and labor economics (NBER; OECD; ILO).

27 | GLOSSARY

Glossary of Key Terms

A compact glossary for consistent use of terms across this survey and any documents derived from it [ISO/IEC 22989; NIST AI RMF].

Agent (AI agent) A system that receives an objective, plans steps, uses tools and may execute actions with limited human intervention.

AI-native organization An organization whose products, operations and decision-making are designed around AI from the start.

AI-orchestrated organization An organization in which agents coordinate information, tasks, decisions and workflows across teams and systems.

Assistant (AI assistant) A system that supports a person by retrieving, summarizing, recommending or drafting, without acting independently.

Autonomy level A graded scale (0–6) describing how much of a task an AI system performs and how much human involvement remains.

Copilot An AI that works inside a business process and suggests actions while the employee remains responsible for the final action.

Data capital The stock of trusted, governed, accessible data an organization can use.

Foundation model A large pretrained model (e.g., an LLM) that can be adapted to many tasks.

Guardrails Technical and organizational controls that constrain AI behavior (permissions, filters, approvals).

Human capital The stock of AI-literate talent, leadership and workforce skills.

Human-in-the-loop A design in which a human approves, reviews or monitors AI actions at defined checkpoints.

LLMOps Operational practices for running large language models in production (routing, cost, evaluation, monitoring).

MLOps Operational practices for the model lifecycle (registry, deployment, monitoring, retraining).

Multi-agent system Several specialized agents that cooperate (e.g., planning, retrieval, analysis, validation, reporting, governance).

Organizational capital The stock of redesigned workflows, incentives, decision rights and collaboration structures.

Proof of concept (PoC) A demonstration that an AI use case can work, before production scaling.

RAG Retrieval-Augmented Generation: grounding model outputs in retrieved, trusted sources with citations.

Shadow AI Use of unapproved AI tools or data outside governance.

Workflow (AI workflow) A predefined, engineer-controlled sequence of AI steps.

APPENDIX | TOOLKIT & RECOMMENDED TITLES

Self-Assessment Toolkit and Recommended Titles

AI transformation scorecard

Score each dimension 1 (foundations) to 6 (AI-native), using the maturity stages in Part 06 and the stocks in Part 13. The lowest score is the ceiling on AI value.

Dimension	Score (1–6)	Evidence to cite
Strategy	___	AI tied to KPIs and an owner
Leadership	___	Sponsorship, AI literacy
Data	___	Quality, access, lineage
Technology	___	Platform, integration, agents
Processes	___	Workflows redesigned
Talent	___	Skills coverage, training
Culture	___	Adoption, experimentation
Governance	___	Risk, audit, approvals
Measurement	___	Baseline, KPIs, outcomes
Customer value	___	P&L and experience impact

Task-level AI decision worksheet

For any candidate task, answer the assistance–replacement matrix from Part 14 and record the recommended autonomy level (0–6).

Question	Answer	If …
Repetitive, low-risk?	___	→ replace/automate
Structured, verifiable?	___	→ automate
Complex, data-rich?	___	→ assist decision
Ambiguous, strategic?	___	→ support, human decides
High-risk, regulated?	___	→ limited assistance
Relationship-sensitive?	___	→ AI prepares, human leads
Safety-critical?	___	→ human controls final action

Recommended additional chapters

The following titles are candidates for the next expansion of this document. Each is grounded in the research base of this survey.

Recommended additional chapters

1. **The Economics of AI Transformation** — ROI by sector, payback periods, TCO and capex/opex allocation [McKinsey 2023; Deloitte].
2. **Sector Deep Dives** — banking, retail, healthcare, manufacturing and public sector [Stan-

ford AI Index; WEF].

3. **AI Product Management Playbook** — discovery, prioritization, human–AI workflows, model and agent requirements [product-management literature].
4. **Agentic AI: From Copilots to Autonomous Workforces** — autonomy levels, orchestration, permissions, escalation [Gartner 2024; MCP].
5. **Responsible AI Metrics that Boards Can Read** — fairness, safety, privacy and audit KPIs [NIST AI RMF; OECD].
6. **The AI-Native Startup Playbook** — building from day one with AI at the core [Stanford AI Index; venture research].
7. **Workforce of the Future: Jobs, Skills and Reskilling at Scale** — labor-market data and training economics [WEF Future of Jobs; OECD].
8. **Measuring Organizational Culture Change** — culture surveys, adoption cohorts and behavioral telemetry [HBR; Microsoft WTI].
9. **AI Transformation in the EU: Regulation as Strategy** — EU AI Act, GDPR, ISO/IEC 42001 as competitive levers [European Commission].
10. **A Quantitative Maturity Index** — a composite AI transformation score combining the stocks, ladder and operating-model assessments.

Source table

Research area	Primary sources	Supporting sources
AI adoption	McKinsey, Deloitte, Stanford	Industry analysis
Transformation models	McKinsey, WEF, Deloitte	Academic research
AI governance	NIST, EU, OECD, ISO	Legal analysis
Workforce	WEF, OECD, Stanford	Academic labor research
Technology	Technical documentation	Engineering publications
Use cases	Company reports	Vendor case studies
Business value	Annual reports, KPI data	Consulting studies
Challenges	Institutional reports	Academic studies

Key takeaway

Use the scorecard and decision worksheet to convert this survey into a working diagnostic for your organization.